Name:

No.:

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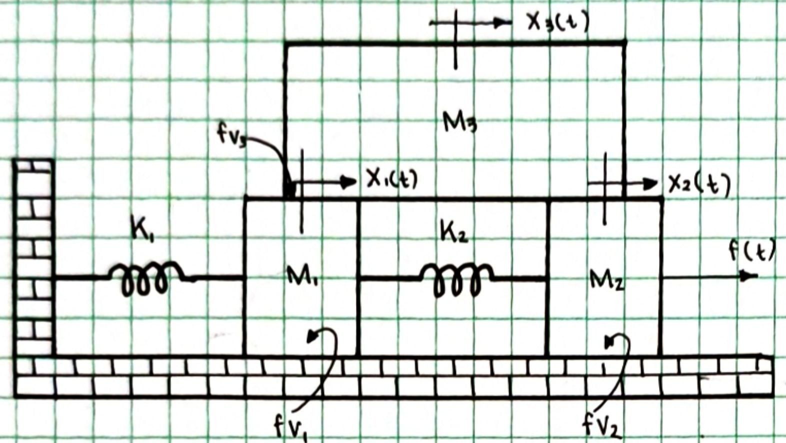
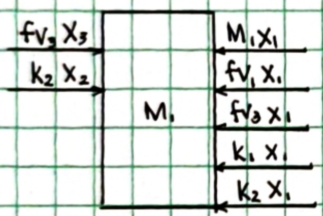
Date:

CONVERT THE FOLLOWING SYSTEM INTO STATE SPACE MODEL AND SIMULATE IN MATLAB USING LIVE SCRIPT AND SIMULINK.

1.





$x_1 = x_1(t)$
$x_2 = \dot{x}_1(t)$
$\dot{x}_1 = \dot{x}_1(t) = x_2$
$\dot{x}_2 = \ddot{x}_1(t)$

$x_3 = x_2(t)$
$x_4 = \dot{x}_2(t)$
$\dot{x}_3 = \dot{x}_2(t) = x_4$
$\dot{x}_4 = \ddot{x}_2(t)$
$u_1 = f(t)$

$x_5 = x_3(t)$
$x_6 = \dot{x}_3(t)$
$\dot{x}_5 = \dot{x}_3(t) = x_6$
$\dot{x}_6 = \ddot{x}_3(t)$

$$0 = M_1\ddot{x}_1 + B_1\dot{x}_1 + B_3\dot{x}_1 + K_1 x_1 + K_2 x_1 - B_3\dot{x}_3 - K_2 x_2$$
$$f(t) = M_2\ddot{x}_2 + B_2\dot{x}_2 + B_4\dot{x}_2 + K_2 x_2 - B_4\dot{x}_3 - K_2 x_1$$
$$0 = M_3\ddot{x}_3 + B_3\dot{x}_3 + B_4\dot{x}_3 - B_1\dot{x}_1 - B_2\dot{x}_2$$

$$0 = M_1 \dot{x}_2 + B_1 x_2 + B_3 x_2 + K_1 x_1 + K_2 x_1 - B_3 x_6 - K_2 x_3$$
$$u_1 = M_2\dot{x}_4 + B_2 x_4 + B_4 x_4 + K_2 x_3 - B_4 x_6 - K_2 x_1$$
$$0 = M_3\dot{x}_6 + B_3 x_6 + B_4 x_6 - B_1 x_2 - B_2 x_4$$

$$\dot{x}_1 = 0x_1 + x_2 + 0x_3 + 0x_4 + 0x_5 + 0x_6 + 0u_1$$
$$\dot{x}_2 = -\left(\frac{K_1+K_2}{M_1}\right)x_1 - \left(\frac{B_1+B_3}{M_1}\right)x_2 + \left(\frac{K_2}{M_1}\right)x_3 + 0x_4 + 0x_5 + \left(\frac{B_3}{M_1}\right)x_6 + 0u_1$$
$$\dot{x}_3 = 0x_1 + 0x_2 + 0x_3 + x_4 + 0x_5 + 0x_6 + 0u_1$$
$$\dot{x}_4 = \left(\frac{K_2}{M_2}\right)x_1 + 0x_2 - \left(\frac{K_2}{M^2}\right)x_3 - \left(\frac{B_2+B_4}{M_2}\right)x_4 - 0x_5 + \left(\frac{B_4}{M_2}\right)x_6 + \frac{u_1}{M_2}$$
$$\dot{x}_5 = 0x_1 + 0x_2 + 0x_3 + 0x_4 + 0x_5 + x_6 + 0u_1$$
$$\dot{x}_6 = 0x_1 + \left(\frac{B_1}{M_3}\right)x_2 - 0x_3 + \left(\frac{B_2}{M_3}\right)x_4 + 0x_5 - \left(\frac{B_3+B_4}{M_3}\right)x_6 + 0u_1$$

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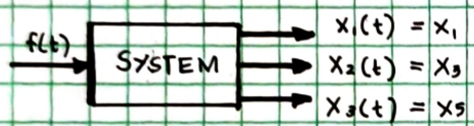
$$
\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \\ \dot{x}_5 \\ \dot{x}_6 \end{bmatrix}
=
\begin{bmatrix}
0 & 1 & 0 & 0 & 0 & 0 \\
-\left(\frac{K_1+K_2}{M_1}\right) & -\left(\frac{B_1+B_3}{M_1}\right) & \left(\frac{K_2}{m_1}\right) & 0 & 0 & \left(\frac{B_3}{M_1}\right) \\
0 & 0 & 0 & 1 & 0 & 0 \\
\left(\frac{K_2}{M_2}\right) & 0 & -\left(\frac{K_2}{M_2}\right) & -\left(\frac{B_2+B_4}{M_2}\right) & 0 & \left(\frac{B_4}{M_2}\right) \\
0 & 0 & 0 & 0 & 0 & 1 \\
0 & \left(\frac{B_1}{M_3}\right) & 0 & \left(\frac{B_2}{M_3}\right) & 0 & -\left(\frac{B_3+B_4}{M_3}\right)
\end{bmatrix}
\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix}
+
\begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{M_2} \\ 0 \\ 0 \end{bmatrix} u_1
$$



$y_1 = x_1 + 0u_1$

$y_2 = x_3 + 0u_1$

$y_3 = x_5 + 0u_1$

$$
y = \begin{array}{c} \begin{matrix} x_1 & x_2 & x_3 & x_4 & x_5 & x_6 \end{matrix} \\
\begin{bmatrix}
1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0
\end{bmatrix} \end{array}
\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix}
+
\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} u_1
$$